

Introduction

INTERMEDIATE CHATGPT



Alex Banks

Founder & Educator

ChatGPT's explosive growth



ChatGPT, developed by OpenAI, was initially released as a prototype in November 2022. It quickly gained attention for its conversational abilities and wide-ranging applicability in generating text-based responses.

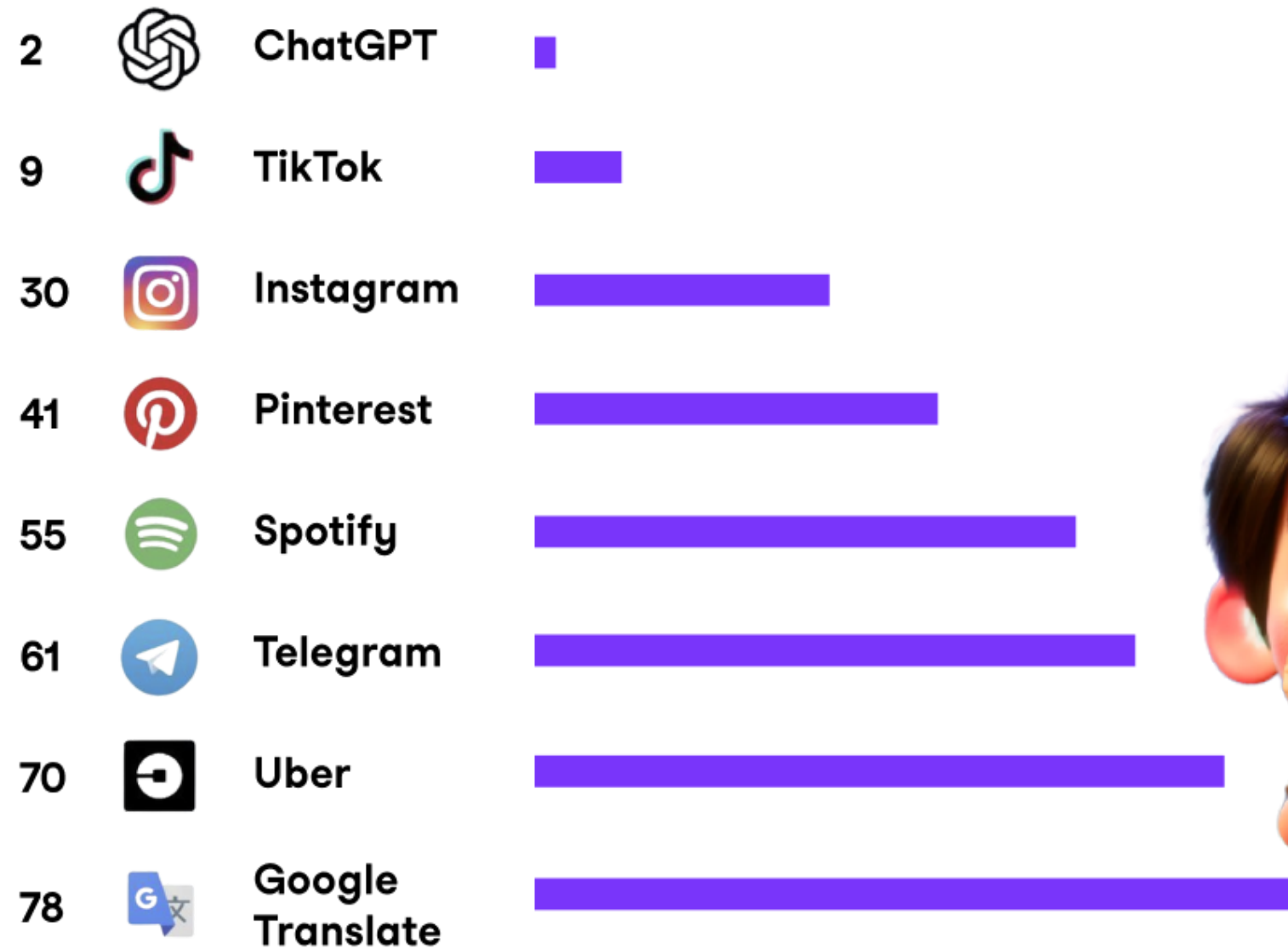
When was ChatGPT released?

5 days → 1M users

¹ Images created using DALLÉ-3

ChatGPT's explosive growth

Months to hit 100M users



ChatGPT's explosive growth



Large Language Models

A primer on Large Language Models (LLMs)

Neural network — predicts the next word in a sequence

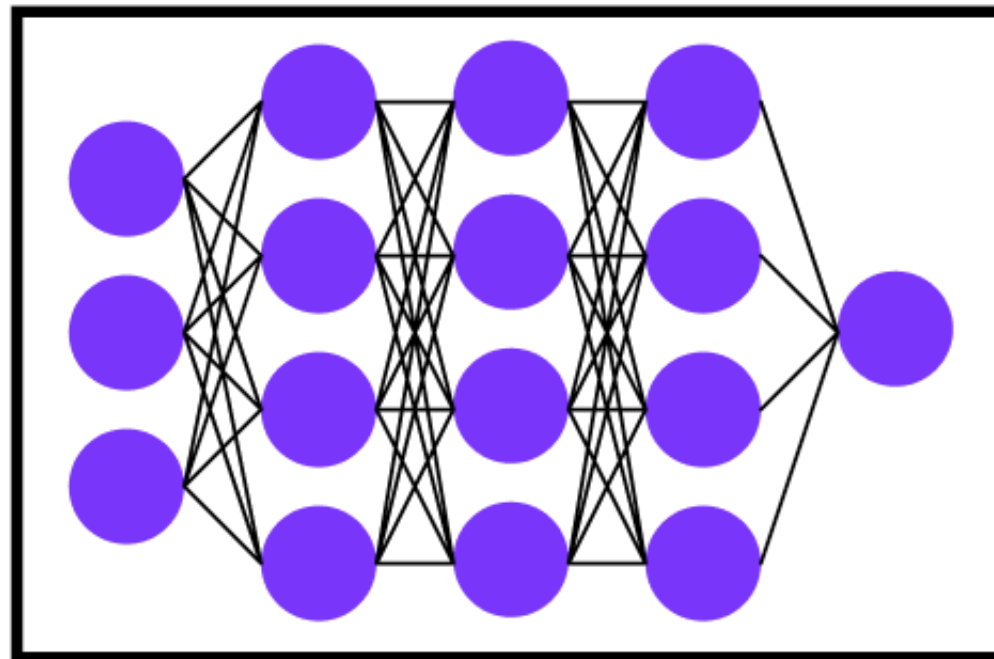


cat →

sat →

on →

a →



→ **mat (97%)**

A primer on Large Language Models (LLMs)

Next token prediction

Jamie likes to eat → **pizza**
Jamie likes to eat → **salad**
Jamie likes to eat → **oatmeal**
Jamie likes to eat → **eggs**



Masked language modeling

Sasha → **loves** → football
Sasha → **hates** → football
Sasha → **enjoys** → football
Sasha → **avoids** → football

The dawn of AI: Recurrent models

Uses hidden states

Process data sequentially



Vanishing gradient problem

Groundbreaking architecture: The transformer

Weighs the significance of words

Attention mechanisms



Process inputs simultaneously

Welcome to Intermediate ChatGPT!

Alex Banks



Welcome to Intermediate ChatGPT

💡 Chapter 1: Technology behind ChatGPT



Welcome to Intermediate ChatGPT

💡 Chapter 1: Technology behind ChatGPT
Chapter 2: Prompt strategies



Welcome to Intermediate ChatGPT

- 💡 Chapter 1: Technology behind ChatGPT
- Chapter 2: Prompt strategies
- Chapter 3: Advanced functions



Let's practice!
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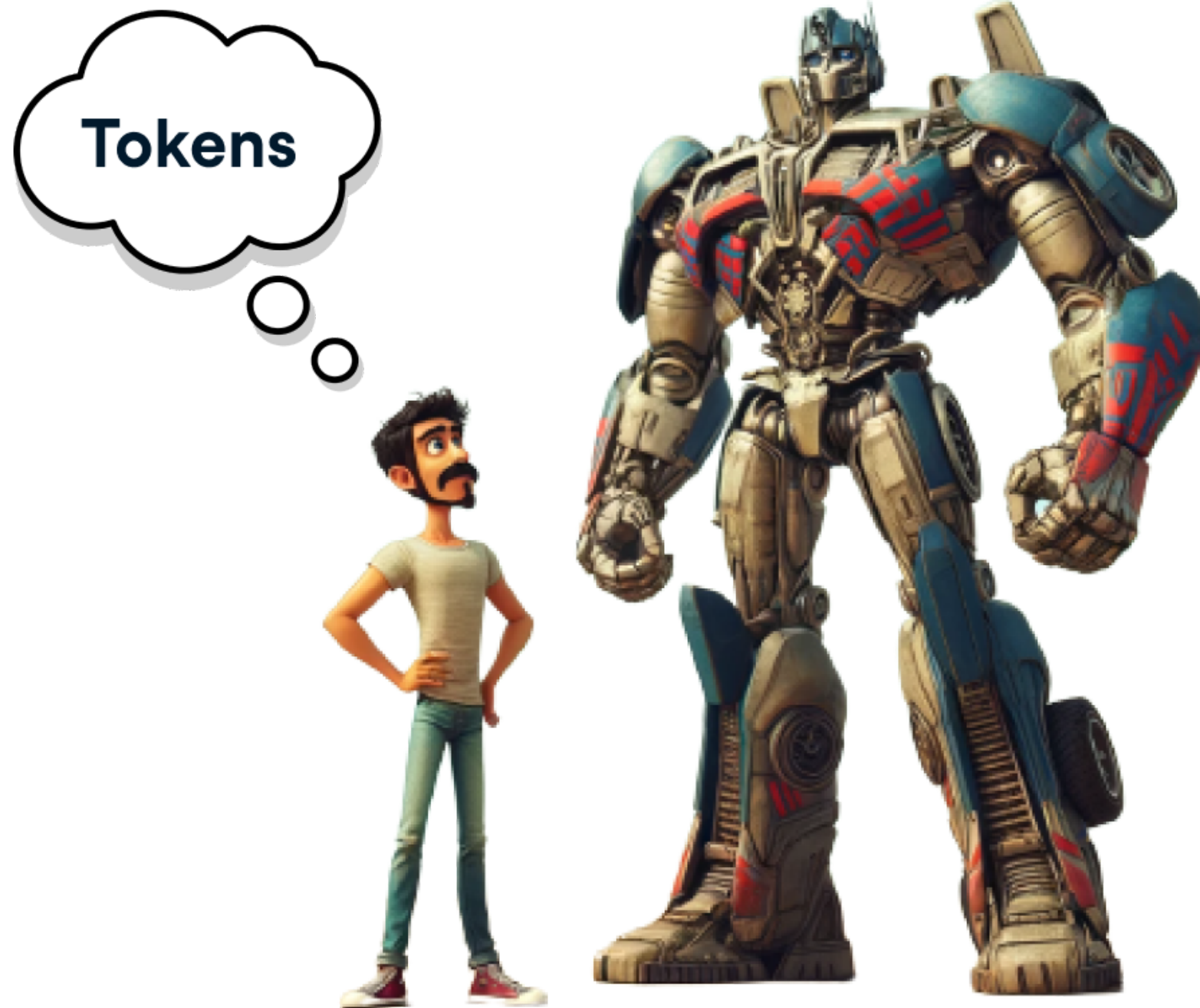
Tokenization and Transformers

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Tokenization

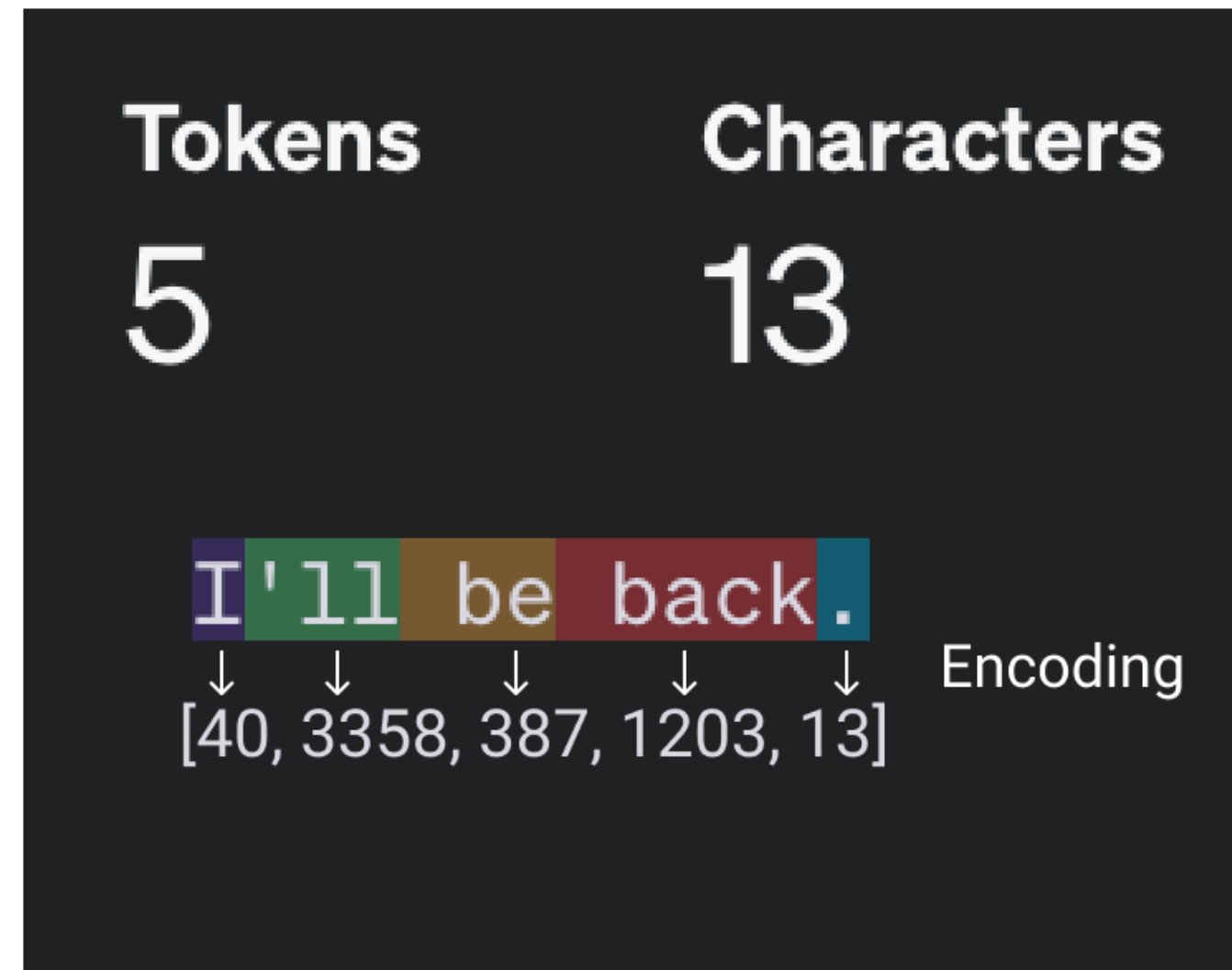


Tokenization

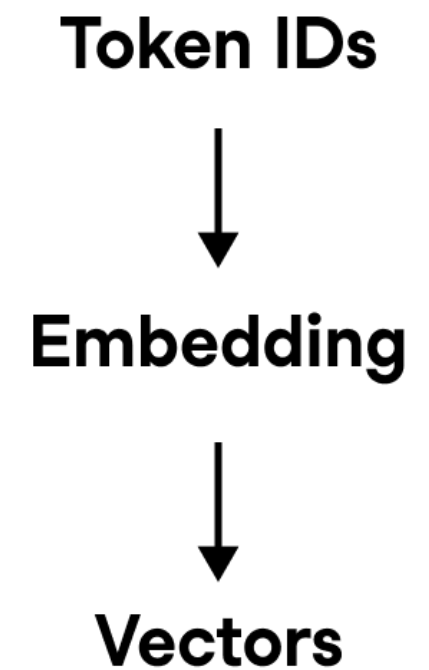
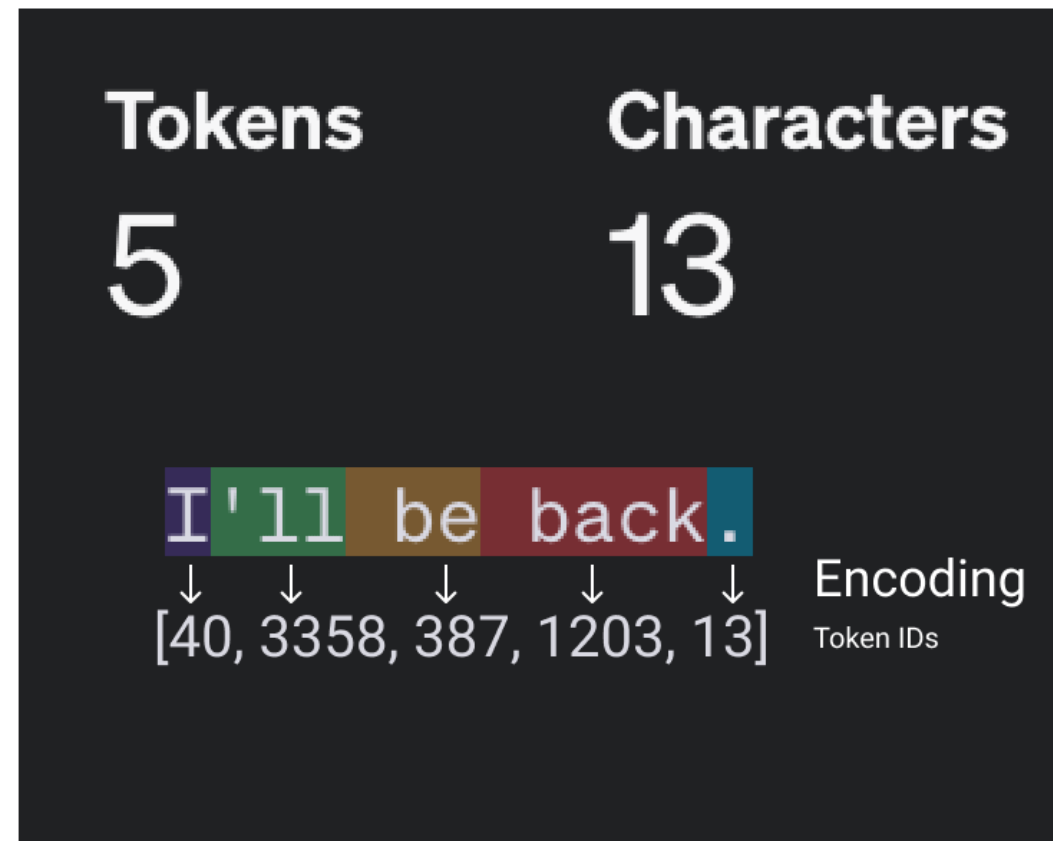


Tokens	Characters
5	13
I'll be back.	

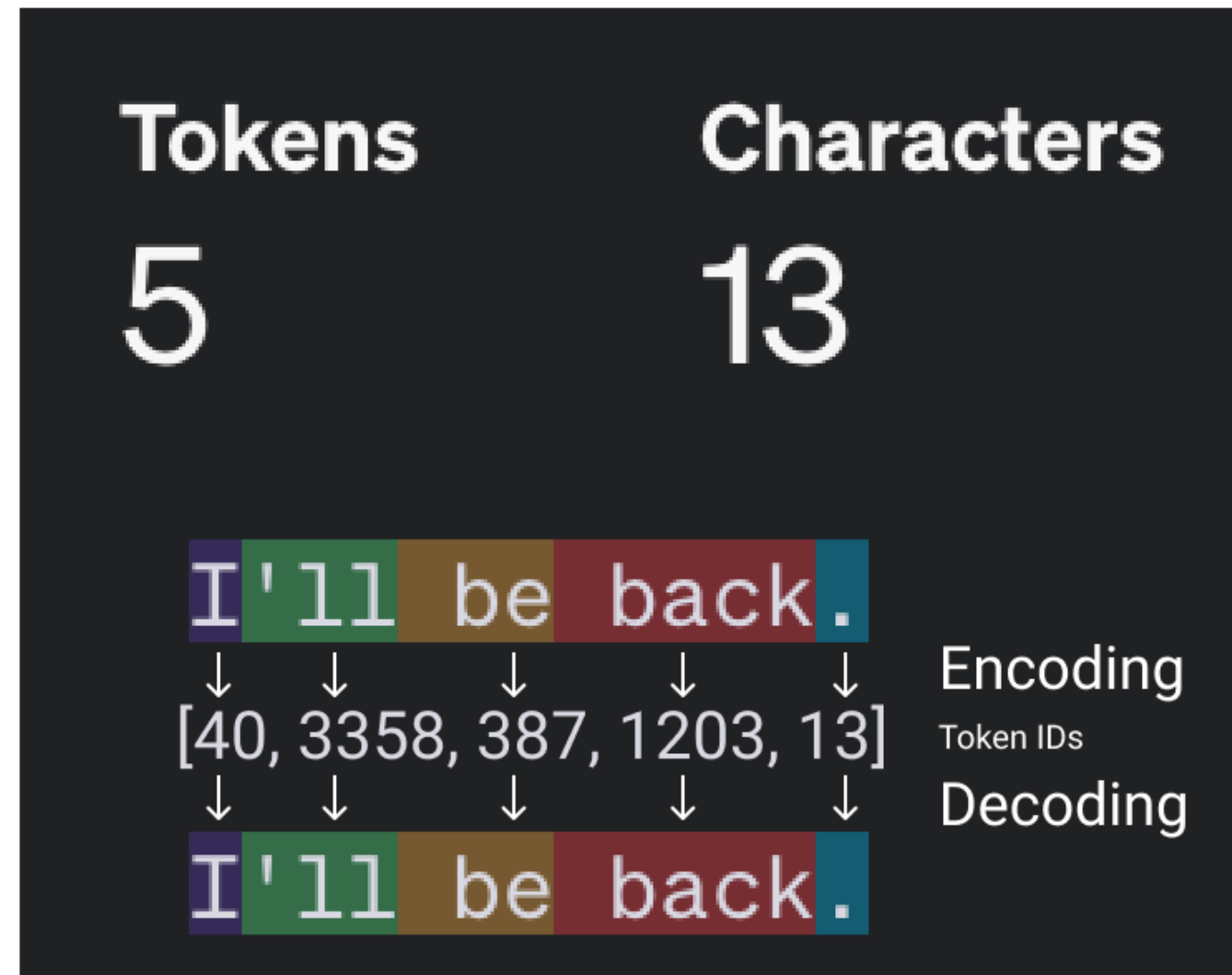
Encoding



Vectorization



Decoding



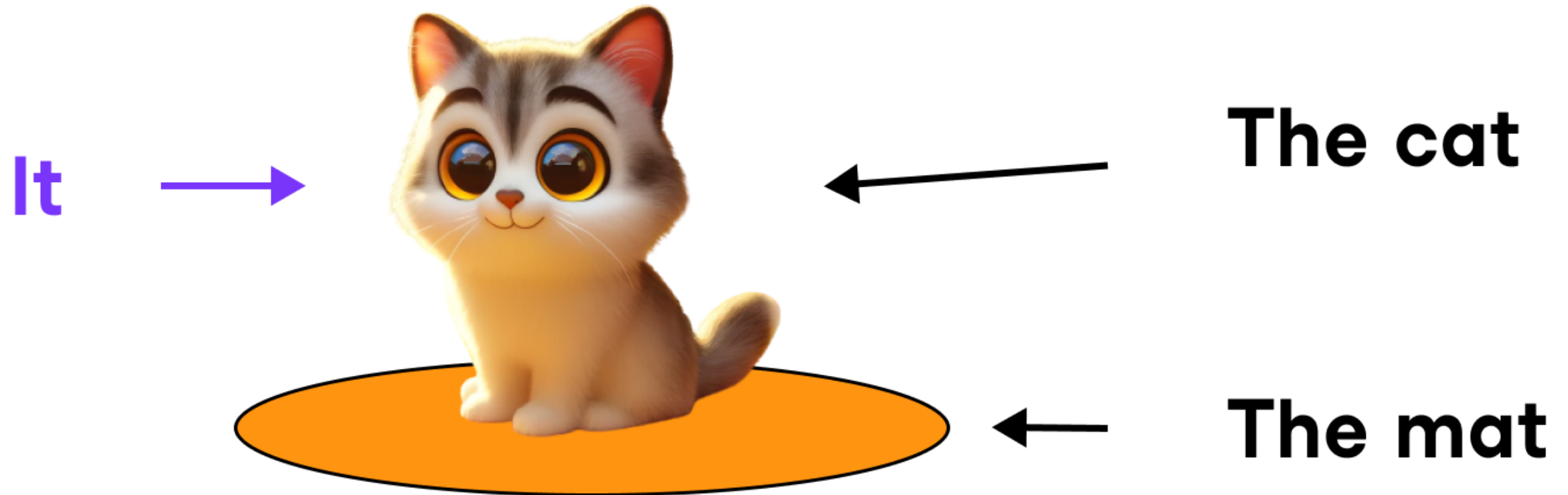
Self-attention

Input



Output

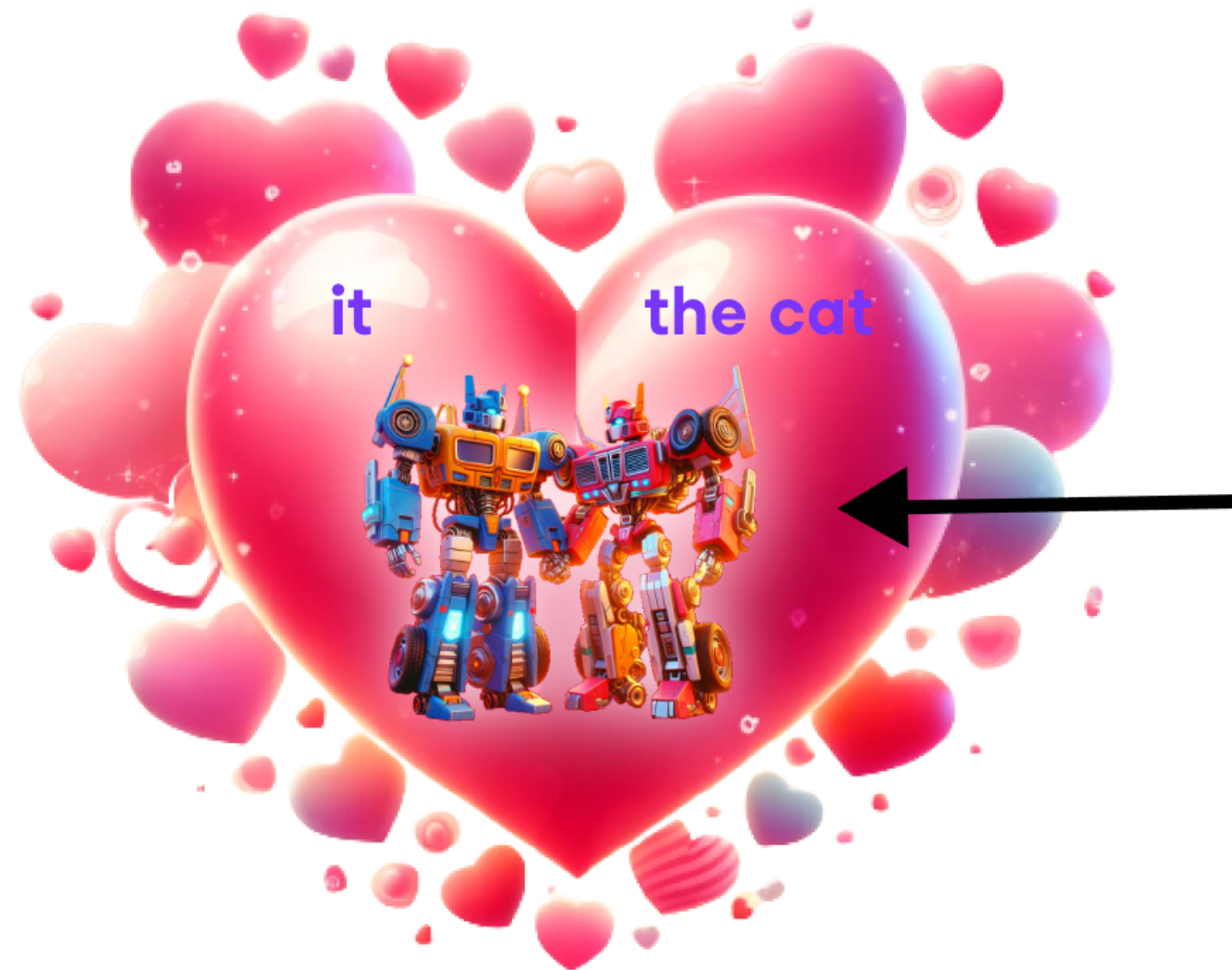
Self-attention



“The cat sat on the mat because *it* was tired.”

Token relationships

"it" → "the cat"
"it" → "sat"
"tired" → "the cat"



**High
attention
score**

Contextual understanding

What's 'it'?

'It' is the cat!



“The cat sat on the mat because **it was tired.”**

Let's practice!
INTERMEDIATE CHATGPT

GPT models and training LLMs

INTERMEDIATE CHATGPT



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The genesis of GPT



**117M
parameters**
GPT-1

**1.5B
parameters**

GPT-2

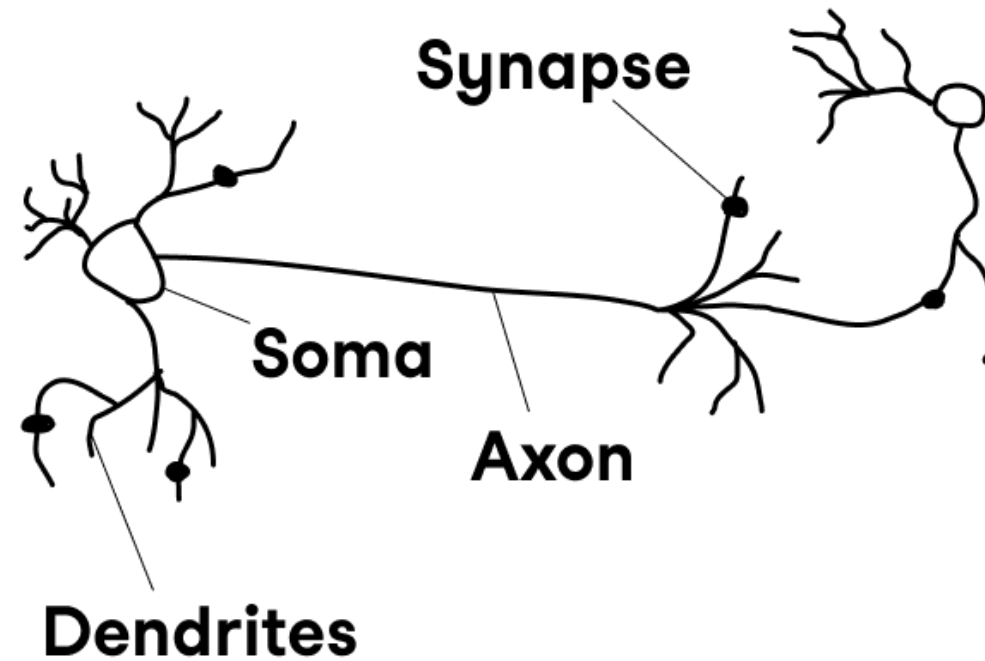
**175B
parameters**

GPT-3

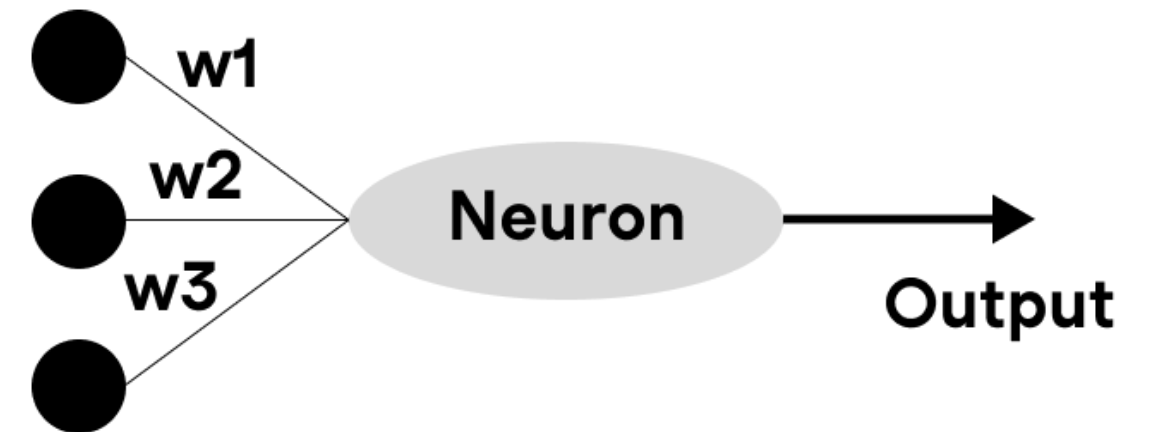
The genesis of GPT



Biological Neural Network



Artificial Neural Network



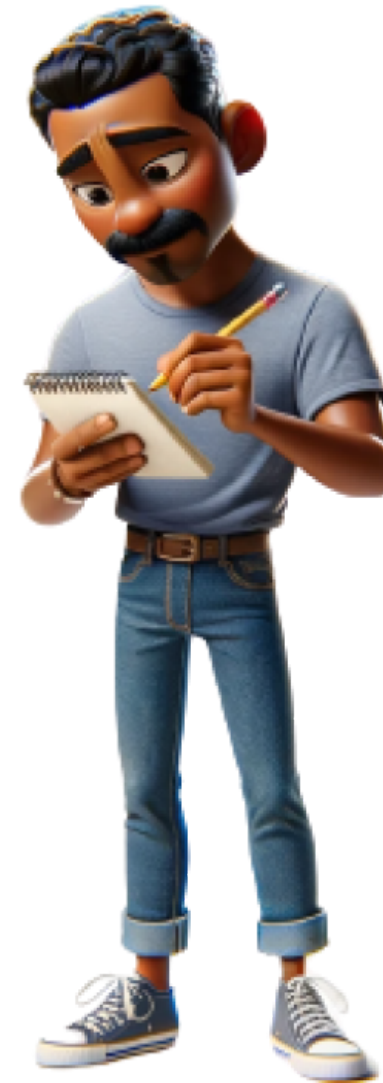
Breaking down GPT



**Generative
Pre-trained
Transformer**

→ **GPT**

Generative





Pre-trained

Transformer





ChatGPT



How do we train LLMs?



\$10M → \$100M+

Step 1 - Pre-training

Large quantities of low-quality internet data



every
~year



Stage 1: Pretraining

1. Download ~10TB of text.
2. Get a cluster of ~6,000 GPUs.
3. Compress the text into a neural network, pay ~\$2M, wait ~12 days.
4. Obtain **base model**.

Model	Release Date	Number of Parameters	Training Dataset Size
GPT-1	2018	117M	8 million web pages
GPT-2	2019	1.5B	40GB of web text
GPT-3	2020	175B	570GB of web text

Step 2 - Fine-tuning

Low quantities of high-quality conversational data

- **Written by people**
- **~100K conversations**

<user>

Can you explain the importance of doing things that don't scale early on in a startup's lifetime. Provide a clear example.

</user>

<assistant>

In the early stages of a startup, doing things that don't scale is crucial for several reasons: 1. Understanding Customer Needs: Engaging directly with your customers helps you understand their needs, preferences, and pain points deeply. This insight is vital for refining your product and ensuring it solves real problems...

</assistant>

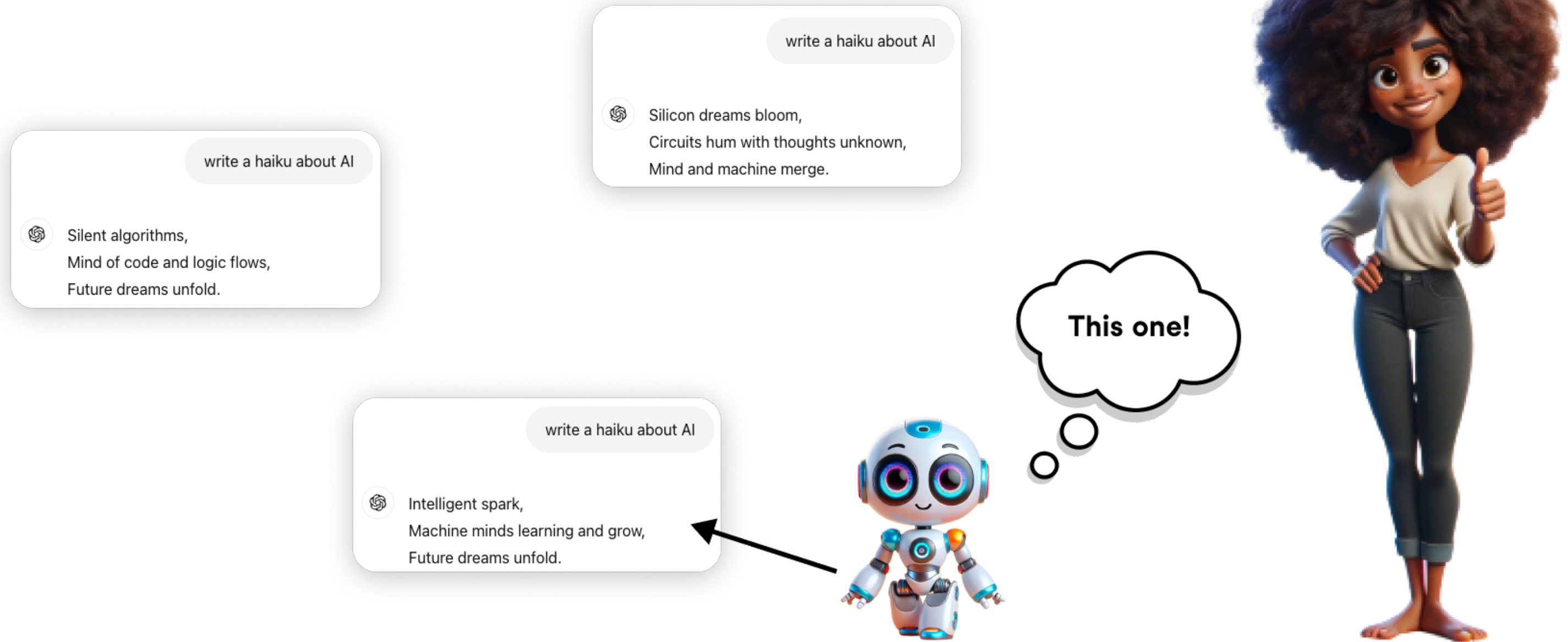


Step 3 - RLHF

RLHF = Reinforcement Learning from Human Feedback

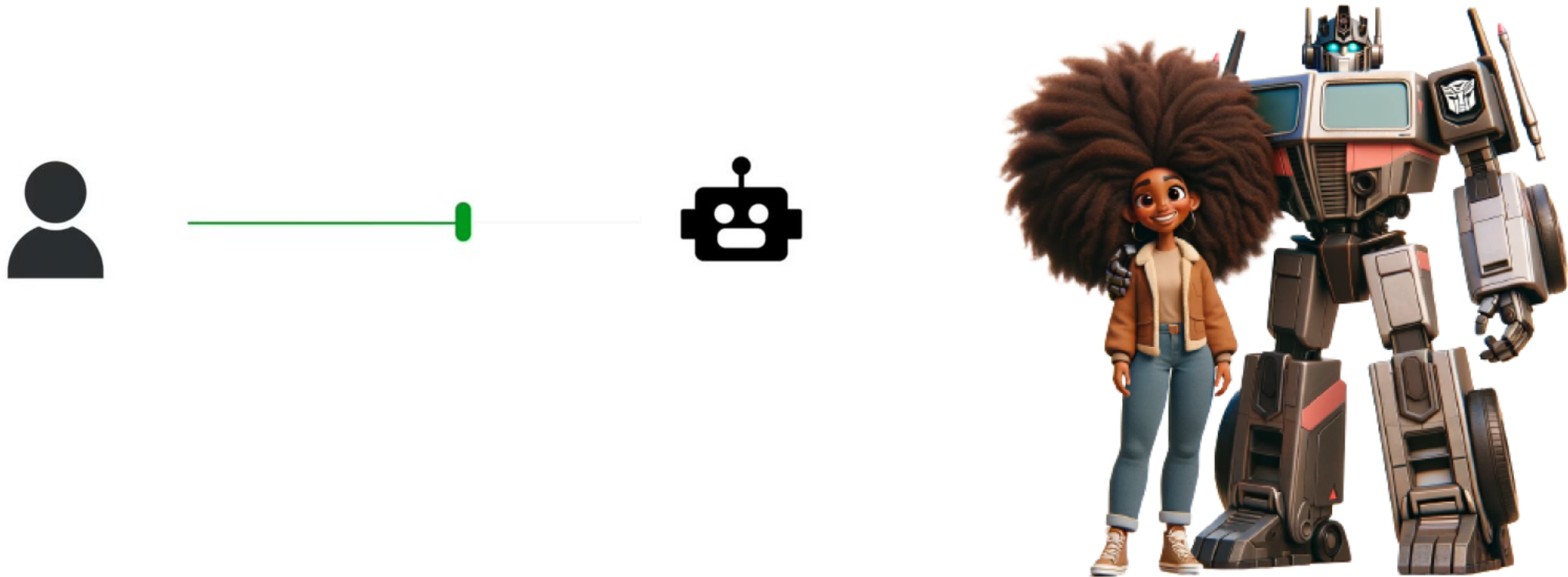


The role of AI in crafting comparison labels



The role of AI in crafting comparison labels

Labeling is a human-machine collaboration



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